A REVIEW OF THE EMPIRICAL DISCLOSURE LITERATURE

Prepared for the 2000 JAE Conference

Paul M. Healy

and

Krishna G. Palepu

Graduate School of Business
Harvard University

December 2000

1. Introduction

Corporate disclosure is critical for the functioning of an efficient capital market.[1] Firms provide disclosure through regulated financial reports, including the financial statements, footnotes, management discussion and analysis, and other regulatory filings. In addition, some firms engage in voluntary communication, such as management forecasts, analysts' presentations and conference calls, press releases, internet sites, and other corporate reports. Finally, there are disclosures about firms by information intermediaries, such as financial analysts, industry experts, and the financial press.

In this paper we review research on financial reporting and voluntary disclosure of information by management, summarize key research findings, and identify areas for future work. Section 2 examines the forces that give rise to demand for voluntary disclosure in a modern capital-market economy, and the institutions that increase the credibility of disclosures. We argue that demand for financial reporting and disclosure arises from information asymmetry and agency conflicts between managers and outside investors. The credibility of management disclosures is enhanced by auditors, standard setters and other capital market intermediaries.

In section 3 we use the disclosure framework to identify important questions for research, and review the questions that researchers have addressed. We conclude that progress has been made in understanding the value of regulated accounting information, the role and effectiveness of intermediaries such as financial analysts, as well as factors

[1] Corporate disclosure can also be directed to stakeholders other than investors. However, there has been relatively little research on these types of voluntary disclosures. Consequently, we focus in this paper on investor communication.

underlying firms' financial reporting and disclosure decisions, and their capital market consequences. However, a number of important questions have yet to be answered. Further, as we discuss in section 4, significant changes in the economic environment have potential to alter the nature of financial reporting and disclosure, creating exciting new opportunities for research.

2. The Role of Disclosure in Capital Markets

In this section we examine the role of disclosure in modern capital markets. Information and incentive problems impede the efficient allocation of resources in a capital market economy. Disclosure and the institutions created to facilitate credible disclosure between managers and investors play an important role in mitigating these problems. This section presents a framework for disclosure and section 3 discusses the implications for research.

A critical challenge for any economy is the optimal allocation of savings to investment opportunities. There are usually many new entrepreneurs and existing companies that would like to attract household savings, which are typically widely distributed, to fund their business ideas. While both savers and entrepreneurs would like to do business with each other, matching savings to business investment opportunities is complicated for at least two reasons. First, entrepreneurs typically have better information than savers about the value of business investment opportunities. Second,

entrepreneurs' communications with investors are not completely credible because investors know entrepreneurs have an incentive to inflate the value of their ideas.

These information differences and conflicting incentives give rise to the "lemons" problem that can potentially lead to a breakdown in the functioning of the capital market.[2] For example, consider a situation where half the business ideas are "good" and the other half are "bad." Both investors and entrepreneurs are rational and value investments conditional on their own information. If investors cannot distinguish between the two types of business ideas, entrepreneurs with "bad" ideas will try to claim that their ideas are as valuable as the "good" ideas. Realizing this possibility, investors will value both good and bad ideas at an average level. Therefore, if the lemons problem is not fully resolved, the capital market will rationally undervalue some good ideas and overvalue some bad ideas relative to the information available to entrepreneurs.

There are several well-known solutions to the lemons problem. Optimal contracts between entrepreneurs and investors will provide incentives for full disclosure of private information, thus eliminating the mis-valuation problem (see Kreps (1990), Chapters 17 and 18). Another potential solution to the information asymmetry problem is regulations that require managers to fully disclose their private information. Finally, the lemons problem can be resolved by effective information intermediaries, such as financial analysts and rating agencies, who engage in private information production to uncover managers' superior information.

[2] See Akerlof (1970).

Figure 1 provides a schematic of the role of disclosure, and information and financial intermediaries in the working of capital markets. The left side of figure 1 presents the flow of capital from savers to firms. Capital can flow to business ideas in two ways. First, it can flow directly from savers to businesses. Examples include private equity financing and angel financing. A second and more typical way for capital to flow from savers to businesses is through financial intermediaries. The right side of the figure presents the flow of information from businesses to savers and intermediaries. Firms can communicate directly with investors through such media as financial reports and press releases. They also communicate with financial intermediaries or through information intermediaries, such as financial analysts.

A variety of economic and institutional factors determine whether contracting, regulation and information intermediaries eliminate information asymmetry, or leave some residual information problem. Examples include the ability to write and enforce optimal contracts, proprietary costs that might make full disclosure costly for investors, regulatory imperfections, and potential incentive problems for intermediaries themselves. Research on corporate disclosure, therefore, focuses on cross-sectional variation in these factors and their economic consequences.[3]

3. Research on Disclosure

[3] A similar approach is used by finance scholars to study corporate finance issues such as capital structure, dividends/stock repurchases and private equity financing (see, for example, Myers and Majluf (1986)).

We use the lemons framework for examining disclosure raises a number of important research questions on disclosure regulations, the effectiveness of auditors and information intermediaries, decisions by managers on disclosure, and the economic consequences of disclosure. Table 1 summarizes these questions. The remainder of this section discusses the findings and limitations of research on these questions, as well as opportunities for future research. We focus on empirical research; analytical research is covered by other papers in this issue (see Verrecchia (2001), Dye (2001), and Lambert (2001)).

3.1 Regulation of Disclosure and Financial Reporting

3.1.1 Regulation of Disclosure

Many economic models assume that demand for regulation arises when there are market imperfections or externalities. Several externalities associated with disclosure are discussed in the literature. Leftwich (1980), Watts and Zimmerman (1986), and Beaver (1998) note that accounting information can be viewed as a public good since existing stockholders implicitly pay for its production but cannot charge potential investors for their use of it. Prospective investors, therefore, free ride on information paid for by existing shareholders, leading to the potential underproduction of information in the economy.

A second explanation for regulation, also discussed by Leftwich (1980), Watts and Zimmerman (1986), and Beaver (1998), proposes that disclosure regulation is motivated by concerns other than market failures. For example, regulators may be concerned about

the welfare of financially unsophisticated investors. By creating minimum disclosure requirements, regulators reduce the information gap between informed and uninformed. However, unsophisticated investors are not prevented from investing in financial knowledge or in hiring the services of sophisticated intermediaries if there is no regulation.

The above arguments for regulation leave many unanswered questions. For example, is the magnitude of potential market failures in disclosure significant? Is regulation of disclosure likely to materially improve the situation? Posner (1974) argues that regulators tend to become captured by those they regulate (see also Watts and Zimmerman (1986)). Is this an important problem for financial reporting? Finally, if regulation is effective in increasing economic efficiency, what types of disclosure should be required by regulation and what should be left to the discretion of management?

Whether a market failure, such as that discussed for disclosure, is corrected through regulation is an empirical question. However, empirical research on the regulation of disclosure is virtually non-existent. This is somewhat surprising given the central role that regulation has played in financial reporting, and the limitations of the economic arguments supporting regulation.

3.1.2 Regulation of Financial Reporting Choices

Accounting standards regulate the reporting choices available to managers in presenting the firm's financial statements. This type of regulation potentially reduces processing costs for users of financial statements by providing a commonly accepted language that managers can use to communicate with investors.

Several questions arise about the regulation of financial reporting methods. First, what are the objectives of standard setters? How do they decide to examine certain reporting issues and not others? Second, what are optimal forms of organization and due process for standard setting bodies? These issues have become highly relevant in the recent debate on the organizational structure of the IASC. Third, do accounting standards add value for investors or other stakeholders?

Accounting research has largely focused on the third question. This research has taken two forms. In the "capital markets" research, studies examine the relation between accounting information and security prices. This research is extensively reviewed by Kothari (2001). The most significant conclusion of this research is that regulated financial reports provide new and relevant information to investors. Further, this research documents that the informativeness of required accounting varies systematically with firm and country characteristics (see Collins and Kothari (1989), Easton and Zmijewski (1992), Alford, Jones, Leftwich and Zmijewski.(1993), and Ball, Kothari and Robin (2000)). Several recent studies document a decline in the level of relevance of earnings and other financial statement items over the last twenty years. Using a variety of different research designs, Chang (1998), Lev and Zarowin (1999), and Brown, Lo, and Lys (1999) find that, in the U.S., the relations between stock returns and earnings, and between stock prices, earnings and book values have deteriorated over time.[4]

[4] Collins, Maydew , and Weiss (1997), Francis and Schipper (1999), and Ely and Waymire (1999) examine the relation between returns, earnings and book values. They conclude that the relation between returns and earnings has deteriorated, but that this has been offset by an increase in the value-relevance of book values. However, Chang (1998) argues that these findings are sensitive to the authors' research design choices.

The above evidence suggests that regulated financial information provides valuable information to investors. However, because this research does not compare the relative informativeness of regulated and unregulated financial information, it does not necessarily imply that regulation is superior to a free market approach to disclosure. The finding that the value of regulated accounting data varies systematically based on firm characteristics, time-dependent variables, and country-specific institutions is also subject to alternative interpretations. Do the differences reflect the influence of systematic economic factors that make regulation more or less effective? Or, is the variation driven by correlated omitted variables such firm- and country-growth, or risk?

Another branch of accounting research examines the value relevance of information presented under proposed new financial reporting standards. This research uses the association between earnings and stock prices or returns as a measure of value relevance. The evidence from this literature indicates that most recent standards generate accounting information that is value relevant. One notable exception identified by researchers is inflation accounting, where no relation to stock prices or returns is observed (see Beaver, Christie and Griffin (1980), Gheyara and Boatsman (1980), and Ro (1980)). A more comprehensive discussion of this literature is provided by Holthausen and Watts (2001) who criticize the use of value relevance as a metric for evaluating accounting standards, and Barth, Beaver and Landsman (2001) who provide an alternative view point on the subject.

We identify four areas where we believe additional research on the role of standard setting is warranted. First, do the current accounting standards provide timely information

to investors, or simply confirm information that is already available to them through other sources? Event study tests of the information content of particular accounting methods attempt to evaluate the timeliness of alternative methods.

Second, studies that evaluate the benefits of alternative reporting methods under consideration by standard setters are more likely to provide useful evidence if they examine costs and benefits of all of the alternatives considered. This has usually been difficult to accomplish since researchers do not have access to the inside information required to estimate performance under different reporting alternatives. Several studies attempt to mitigate this problem. Barth (1991) evaluates alternative metrics for pension liabilities using public disclosures available prior to the release of the new accounting standard. Healy, Myers and Howe (1999) develop a simulation model for a pharmaceutical firm, allowing them to compare the value relevance of alternative measures of R&D performance.

A third area for future research in the standard setting arena is to assess which types of standards are likely to be most useful for investors and other stakeholders. For example, on average do bright-line rules produce information that is more useful than rules that require managers to exercise judgment in selecting reporting methods? Under what conditions do bright-line rules dominate those that require management judgment, and vice versa?

Our final area for future research on standard setting is to examine optimal standards across countries. There is currently heated debate about the merits of global standards. Global standards, however, are only likely to be optimal if the institutions that

monitor and enforce adherence to standards work equally well across countries. In the U.S., auditors provide assurance that the financial statements comply with accounting regulations, and the SEC has enforcement authority. Dechow, Hutton, and Sloan (1996) and Beneish (1999) find that U.S. companies face a significant stock price penalty if the SEC decides to pursue them for violating accounting standards. However, the magnitude of type one and type two errors in SEC enforcement actions is unclear. The effectiveness of monitoring and the penalties from enforcement of standards in other countries, particularly those that are only beginning to develop capital markets, is even more open to question.[5]

In summary, surprisingly little is known about why disclosure is regulated in the capital market. Is there a significant market imperfection or externality that regulation attempts to resolve? If so, how effective is disclosure regulation in resolving this problem?

3.2 Role of Auditors and Intermediaries in the Disclosure Process

3.2.1 Auditors

Auditors provide investors with independent assurance that the firm's financial statements conform to GAAP. The fact that stock prices react to earnings announcements (see Kothari (2001)) suggests that overall investors regard accounting information as credible. However, it is unclear whether the credibility arises from the assurance provided

[5] See DeFond, Wong, and Li (1999), Ball, Kothari and Robin (2000), Ball, Robin and Wu (2000.), and Eccher and Healy (2000) for studies that begin to examine this issue.

by the auditor or from other sources, such as managers' potential legal liability for providing misleading disclosures.

Studies of audit effectiveness examine whether audit qualifications add value for investors and whether auditors' actions are independent of the interests of their clients. Research shows that capital providers require firms to hire an independent auditor as a condition of financing, even when it is not required by regulation. For example, Leftwich (1983) finds that banks require firms to present audited financial information, even for private companies. This implies that capital providers regard auditors as enhancing credibility.

To our knowledge, there is no research that examines directly whether or not auditors significantly enhance the credibility of reported financial statements. Available evidence suggests that auditor qualifications do not provide timely signals to the capital markets. For example, studies of the stock market reaction to audit qualifications show that qualified opinions do not provide new information to investors, in part because they can be anticipated (see Dodd, Dopuch, Holthausen and Leftwich (1984) Dodd, Holthausen and Leftwich (1986), and Dopuch, Holthausen and Leftwich (1986)). This evidence suggests that audit qualifications at best confirm information already available to investors. Choi and Jeter (1992) report that subsequent to qualifications, firms show lower stock price responses to earnings. However, since the study does not control for the unusual performance of firms with qualifications, it is difficult to attribute the decline in earnings response coefficients to reduced credibility.

There are several potential explanations for the paucity of evidence on the value of auditor opinions to investors. Watts and Zimmerman (1981a, 1981b) posit that auditors act

in the interest of the managers that hire them, rather than in the interest of the firms' investors. They report evidence that is weakly consistent with this hypothesis using data on auditors' responses to proposed new accounting standards. An alternative explanation is that auditors provide formal assurance only on the annual report, making it difficult for them to provide timely signals to the capital markets. A third explanation is that auditors are concerned about minimizing their legal liability, rather than enhancing the credibility of financial reports. Accordingly, they lobby for standards that reduce their own risk, even though such standards reduce the value of financial reports to investors. Future research may be able to distinguish between these explanations.

Several additional questions arise about the value of audit opinions. First, how do consulting services provided to audit clients affect auditors' perceived and actual independence, and the value of their audit report? Several of the large audit firms have recently spun-off their consulting operations, providing an opportunity to examine whether their clients have more credible financial statements than those for clients of competitors that continue to provide both audit and consulting services.

Second, there have been a number of important changes in the audit environment in the late 1990s. Legal and organizational changes have limited auditors' liability for audit failures. Also, several of the large audit firms have recently made significant changes in their audit methodology, focusing on a business audit rather than a transactions audit. What impact have these changes had on audit failures and the credibility of financial statements?

Third, what factors influence the credibility of audit reports and financial statements across countries? Factors that are likely to affect the credibility could include differences in audit standards and differences in professional training requirements. DeFond, Wong and Li (1999) examine the effect of new auditing standards that improved auditor independence in China. They find that the new standards increased the frequency of qualified opinions, but that this was accompanied by a 'flight from audit quality.' However, in general the role

of auditors and auditing standards in emerging markets has been unexplored in the literature.

3.2.2 Intermediaries

Studies of the value of intermediaries largely focus on financial analysts. Financial analysts collect information from public and private sources, evaluate the current performance of firms that they follow, make forecasts about their future prospects, and recommend that investors buy, hold or sell the stock. Academic studies focus on information provided to investors from two summary measures produced by analysts, earnings forecasts and buy/hold/sell recommendations. Overall, this evidence indicates that financial analysts add value in the capital market. Their earnings forecasts are more accurate than time-series models of earnings, presumably in part because they are able to incorporate more timely firm and economy news into their forecasts than time-series models (see Brown and Rozeff (1978), Brown, Griffin, Hagerman and Zmijewski (1987), and Givoly (1982). Also, analysts' earnings forecasts and recommendations affect stock prices (see Givoly and Lakonishok (1979), Lys and Sohn (1990), and Francis and Soffer (1997)).

There is also evidence of analyst bias in forecasting and making recommendations. Early evidence on bias indicated that analyst earnings forecasts tended to be optimistic, and that their recommendations were almost exclusively for buys (see Brown, Foster and Noreen (1985)). However, recent evidence indicates a change in the pattern of analysts'

earnings forecasts in the late 1990s. In this period there has been a marked decline in analyst optimism (see Brown (1997) and Matsumoto (2000)).

Research on the role of financial analysts in capital markets indicates that they play a valuable role in improving market efficiency. For example, Barth and Hutton (2000) find that stock prices for firms with higher analyst following more rapidly incorporate information on accruals and cash flows than prices of less followed firms.

Recent research on analysts attempts to improve our understanding of their cross-sectional performance. Two factors are examined, cross-sectional variation in analysts' incentives and expertise. Studies on incentives note that analysts are rewarded for providing information that generates trading volume and investment banking fees for their brokerage houses. Thus, analysts have incentives to make optimistic forecasts and recommendations when their brokerage house has been hired to underwrite or is being considered to underwrite a new securities issue (see Lin and McNichols (1998) and Dechow, Hutton and Sloan (2000)).

Studies of the role of analysts' expertise examine factors likely to influence their aptitude, such as experience, brokerage affiliation, and company or industry assignments. Jacob, Lys and Neale (1999) find that analyst forecast accuracy is affected by innate ability, company assignments, brokerage affiliation, and industry specialization. There appears to be little benefit from experience. Gilson, Healy, Noe and Palepu (2000) find that, for focused companies, analysts that specialize by industry issue more precise forecasts than non-specialist analysts.

Academic research on financial analysts also examines whether there is any relation between management's disclosure decisions, and analyst decisions to cover firms. Bhushan (1989a, 1989b) and Lang and Lundholm (1993) argue that voluntary disclosure lowers the cost of information acquisition for analysts and hence increases their supply. However, the effect of voluntary disclosure on demand for analysts' services is ambiguous. Expanded disclosure potentially enables financial analysts to create valuable new information, such as superior forecasts and buy/sell recommendations, thereby increasing demand for their services. However, public voluntary disclosure also pre-empts analysts' ability to distribute managers' private information to investors, leading to a decline in demand for their services.

Lang and Lundholm (1993) find that firms with more informative disclosures have larger analyst following, less dispersion in analyst forecasts, and less volatility in forecast revisions. Healy, Palepu and Hutton (1999) show that firms with increased analyst ratings of disclosure have lower analyst coverage than their industry peers in the pre-event period. After the increase in disclosure, however, analyst coverage for the sample firms reverts to the same level as other firms in the industry. Finally, Francis, Hanna and Philbrick (1998) find that there is an increase in analyst coverage for firms making conference calls.

Studies of intermediaries other than financial analysts include tests of the value provided by business journalists that analyze and evaluate companies' financial reporting decisions, and bond-rating agencies. Foster (1979, 1987) examined stock price reactions to the publication by Barrons of articles by Abraham Briloff, an academic accountant who periodically questioned firms' accounting decisions. Foster found that firms whose

accounting was challenged by Briloff on average suffered an 8% decline in stock price when the article was released. He concluded that this reaction probably reflected Briloff's superior insights and analysis.

Studies of the value provided by bond-rating agencies (see Holthausen and Leftwich (1986) and Hand, Holthausen and Leftwich (1992)) conclude that rating downgrades provide new information to investors, but that upgrades are already reflected in stock and bond prices when they are announced.

In summary, there has been considerable academic research on the value provided by auditors and financial intermediaries in reviewing firm's disclosures and in making their own disclosures on the firm. This evidence shows that at least some of the disclosures made by auditors, financial analysts, the business press, and bond-rating agencies affect stock prices. However, there remain important gaps in our knowledge about the incentives of auditors and intermediaries, and the impact on their credibility.

3.3 Managers' Reporting Decisions

Research on managers' reporting decisions has focused on two areas. The first area, often called positive accounting theory, focuses on management's financial reporting choices. We provide a brief review of this literature; a more comprehensive survey is provided by Fields, Lys and Vincent (2001). The second area, the voluntary disclosure literature, focuses on management disclosure decisions.

3.3.1 Positive Accounting Theory Literature

The positive accounting theory literature has studied management's motives for making accounting choices and voluntary disclosures when markets are semi-strong form efficient, there are significant costs in writing and enforcing contracts, and there are political costs arising out of the regulatory process (see Watts and Zimmerman (1978 and 1986)). The central focus of this literature is to examine the role of contracting and political considerations in explaining management accounting choices.[6] Two types of contracts are examined, contracts between the firm and its creditors (debt contracts), and contracts between management and shareholders (compensation contracts). Political considerations include management's concern about attracting explicit or implicit taxes, or regulatory actions.

Empirical studies of positive accounting theory test whether managers make accounting method changes or accrual estimates to reduce the costs of violating bond covenants written in terms of accounting numbers, to increase the value of earnings-based bonuses under compensation contracts, or to reduce the likelihood of implicit or explicit taxes. Findings indicate that firms that use accounting methods to accelerate earnings are small and have relatively high leverage. Also, firms' accrual decisions appear to be affected by compensation contracts.

While a majority of positive accounting studies focused on analyzing post-contracting opportunistic accounting choices, some studies view the choice of accounting

[6] Contracts are not the only mechanisms for dealing with information asymmetry discussed in the positive accounting literature. For example, Watts and Zimmerman (1983, 1986) discuss the role of reputation as a mechanism for resolving information problems in the context of auditing.

and disclosure as part of the contracting process itself. Holthausen and Leftwich (1983), Watts and Zimmerman (1990), Smith and Watts (1992), and Skinner (1993) argue that the use of accounting information in lending and compensation contracts should be viewed as endogenous. Consequently, the nature of a firm's assets and its investment opportunity set simultaneously determine its optimal contracting relations and its accounting method choices. Watts and Zimmerman (1983) examine the role of voluntary interim reporting as an *ex ante* contracting part of corporate governance. The *ex ante* role of accounting in the contracting process is also examined by Zimmer (1986), Christie and Zimmerman (1994), and Skinner (1993).

Although positive accounting theory studies generated several interesting empirical regularities regarding firms' accounting decisions, a number of questions arose about how to interpret their evidence (see reviews by Holthausen and Leftwich (1983) and Watts and Zimmerman (1993)). For example, size is typically viewed as a proxy for political sensitivity, but is likely to proxy for many other factors. Also, accounting decisions by managers of highly leveraged firms in financial distress may in part reflect an attempt to conserve cash, or changes in investment opportunities.[7]

3.3.2 Voluntary Disclosure Literature

Research on voluntary disclosure focused on the information role of financial reporting for capital markets (see Healy and Palepu (1993 and 1995)). This research

[7] Palepu (1987), Healy and Palepu (1990), and DeAngelo, DeAngelo and Skinner (1996) report evidence consistent with this hypothesis.

supplemented the positive accounting literature by focusing on stock market motives for accounting and disclosure decisions.

Disclosure studies assumed that managers have superior information to outside investors on their firms' expected future performance. If auditing and accounting regulations work perfectly, managers' accounting decisions and disclosures communicate changes in their firm's business economics to outside investors. Alternatively, if accounting regulation and auditing are imperfect, a more likely possibility, managers trade off between making accounting decisions and disclosures to communicate their superior knowledge of firm's performance to investors, and to manage reported performance for contracting, political or corporate governance reasons. Management motives for making voluntary disclosure and their credibility are, therefore, interesting empirical questions. Below, we discuss empirical evidence on these questions. We also separately analyze potential limitations of the research, many of which are shared across studies.

Motives for Voluntary Disclosure

Researchers discuss six forces that affect managers' disclosure decisions for capital market reasons: capital market transactions, corporate control contests, stock compensation, litigation, proprietary costs, and management talent signaling.

1. Capital markets transactions hypothesis

Healy and Palepu (1993, 1995) hypothesize that investors' perceptions of a firm are important to corporate managers expecting to issue public debt or equity or to acquire

another company in a stock transaction. Consider a firm whose mangers have superior information relative to outside investors regarding the firm's future prospects. Myers and Majluf (1986) point out that if this information asymmetry cannot be resolved, such firms will view making public equity or debt offers to be costly for existing shareholders. Consequently, managers who anticipate making capital market transactions have incentives to provide voluntary disclosure to reduce the information asymmetry problem, thereby reducing the firm's cost of external financing.

A similar conclusion can be drawn from studies that hypothesize that investors demand a risk premium for bearing information risk when there is an information asymmetry between managers and outside investors (see Barry and Brown (1985, 1986) and Merton (1987)). Managers who seek to reduce their cost of capital can do so by reducing information risk through increased voluntary disclosure.

Several studies provide evidence on voluntary disclosure policies of firms issuing new capital. In a comprehensive study of corporate disclosure, Lang and Lundholm (1993) document that analysts' ratings of disclosures are higher for firms issuing securities in the current or future periods. In a subsequent paper, Lang and Lundholm (1997) analyze disclosures specifically for firms that make equity offerings and find that there is a significant increase in disclosure beginning six months before the offering, particularly for the categories of disclosure over which firms have the most discretion. Finally, Healy, Hutton, and Palepu (1999) find that firms with increased analyst ratings of disclosures have an abnormally high frequency of subsequent public debt offers.[8] However, as discussed

[8] Frankel, McNichols and Wilson (1995) find that firms that raise new capital are not more likely to provide management forecasts in the period immediately prior to the offering than at other times. However, this finding is not surprising given that securities laws restrict managers from making forward-looking statements prior to equity offerings.

below, debt and equity offers are not isolated events, making it difficult to assess whether managers' disclosure strategies are caused by public capital market transactions or by omitted related factors.

2. Corporate control contest hypothesis

This hypothesis is motivated by evidence that boards of directors and investors hold managers accountable for current stock performance. Warner, Watts and Wruck (1988), and Weisbach (1988) show that CEO turnover is associated with poor stock performance. Poor stock price performance is also associated with the probability of hostile takeovers, which results in high CEO turnover [see Palepu (1986), Morck, Schleifer, and Vishny (1990)]. DeAngelo (1988) finds that dissident shareholders who wage a proxy fight for board representation frequently cite poor earnings performance as justification for proposed management changes. Voluntary disclosure theory hypothesizes that managers use corporate disclosures to reduce the likelihood of undervaluation and to justify poor earnings performance.

One limitation is that this analysis does not take account of multi-period considerations. For example, if managers expect that a commitment to provide extensive disclosure today could be used to hold them more accountable for any subsequent poor performance, managers of firms subject to corporate control actions may not wish to expand disclosure in a period of poor performance.

There has been relatively little research on voluntary disclosures accompanying hostile takeovers or for firms engaged in proxy contests. One recent exception is Brennan (1999), who finds that targets are more likely to make management earnings forecasts during contested takeover bids.

3. Stock compensation hypothesis

Managers are also directly rewarded using a variety of stock-based compensation plans, such as stock option grants, and stock appreciation rights. These types of compensation schemes provide incentives for mangers to engage in voluntary disclosures for several reasons.

First, managers interested in trading their stock holdings have incentives to disclose private information to meet restrictions imposed by insider trading rules and to increase liquidity of the firm's stock. These types of disclosures should therefore really be classified as required, rather than voluntary, disclosures. Restrictions on insider trading also provide managers with incentives to make voluntary disclosures to correct any perceived undervaluation (relative to their own information set) prior to the expiration of stock option awards.[9]

Second, managers acting in the interests of existing shareholders have incentives to provide voluntary disclosures to reduce contracting costs associated with stock compensation for new employees. Stock compensation is more likely to be an efficient form of remuneration for managers and owners if stock prices are a precise estimate of firm values. Otherwise, managers will demand additional compensation to reward them for bearing any risk associated with misvaluation. Firms that use stock compensation

[9] In the absence of insider trading restrictions, managers may benefit from the undervaluation by buying shares rather than making disclosures to enhance the value of their stock options.

extensively are therefore likely to provide additional disclosure to reduce the risk of misvaluation.[10]

Consistent with this hypothesis, Noe (1999) provides evidence that the incidence of management forecasts is positively associated with trading by insiders in the firm's stock. Aboody and Kasznik (2000) find that firms delay disclosure of good news and accelerate the release of bad news prior to stock option award periods, consistent with managers making disclosure decisions to increase stock-based compensation. Miller and Piotroski (2000) find that managers of firms in turnaround situations are more likely to provide earnings forecasts if they have higher stock option compensation at risk.

4. Litigation cost hypothesis

The threat of shareholder litigation can have two effects on managers' disclosure decisions. First, legal actions against managers for inadequate or untimely disclosures can encourage firms to increase voluntary disclosure. For example, Skinner (1994) hypothesizes that managers of firms with bad earnings news have an incentive to pre-disclose that information to reduce the cost of litigation. This hypothesis presumes that in the absence of litigation managers have an incentive to time the disclosure of good and bad news symmetrically. Litigants and courts, therefore, rationally focus on whether there were delays in bad news announcements.

[10] As discussed in Section 2, we use the term misvaluation to refer to the gap between the value of the firm conditional on managers' information set and on investors' information set. This gap arises when there is information asymmetry between managers and investors that is not fully resolved. Through out our analysis, we assume that both managers and investors are rational, and that stock prices fully incorporate all public information.

One question that arises about the litigation hypothesis is why pre-disclosure of poor performance reduces the risk of litigation. Is it because delaying bad news until a *required* earnings announcement is prima facie evidence that management did not *voluntarily* disclose information to investors in a timely manner? Alternatively, some suggest that pre-disclosure of bad news is beneficial because it spreads the stock price decline over multiple dates, thereby reducing the likelihood of being detected in screens used to identify claims. Of course, this presumes that investors do not make an unbiased assessment of the bad news conveyed by a pre-announcement of earnings. The price drop would then occur at the pre-release date rather than the subsequent earnings announcement, and would continue to hit the screen used to identify potential claims.

In addition, litigation can potentially reduce managers' incentives to provide disclosure, particularly of forward-looking information. This is likely to arise if managers believe that the legal system penalizes forecasts made in good faith because it cannot effectively distinguish between unexpected forecast errors due to chance and those due to deliberate management bias.

The empirical evidence on the litigation hypothesis is mixed. Skinner (1994, 1997) finds that firms with bad earnings news are more than twice as likely to predisclose the poor earnings performance than are firms with good news. In addition, firms with negative earnings news are more likely to be subject to litigation. Finally, he finds weak evidence that litigation costs are lower for firms that predisclose earnings than for those that do not.

In contrast, Francis, Philbrick and Schipper (1994) find that 62% of the firms in their litigation sample were sued over earnings forecasts or preemptive earnings disclosures. In contrast, 87% of their sample of no-litigation firms with comparable stock price declines preannounced an earnings decline. They concluded that predisclosure does not appear to be a deterrent to litigation.

Empirical evidence also suggests that litigation risk is not just relevant for firms with bad news, but also those with good news. For example, Miller and Piotroski (2000) report that managers of turnaround firms in industries subject to high litigation risk are more likely to make management forecasts of positive future earnings information than firms in low risk industries. However, this conclusion should be interpreted with caution because the study only examines the forecasting behavior of firms that ex post experienced a turnaround. It is unclear whether the findings can be extended to firms that were ex ante expected by managers to show a turn around, but failed to do so ex post.

5. Management talent signaling hypothesis

Trueman (1986) argues that talented managers have an incentive to make voluntary earnings forecasts to reveal their type. A firm's market value is a function of investors' perceptions of its managers' ability to anticipate and respond to future changes in the firm's economic environment. The earlier that investors infer that the manager has received information, the more favorable will be their assessment of the manager's ability to anticipate future changes and the higher will be the firm's market value.

6. Proprietary cost hypothesis

Several researchers hypothesize that firms' decisions to disclose information to investors is influenced by concern that such disclosures can damage their competitive position in product markets. [see Verrecchia (1983), Darrough and Stoughton (1990), Wagenhofer (1990), Feltham and Xie (1992), Newman and Sansing (1993), Darrough (1993), and Gigler (1994)]. These studies conclude that firms have an incentive not to disclose information that will reduce their competitive position, even if it makes it more costly to raise additional equity. However, this incentive appears to be sensitive to the nature of the competition, in particular whether firms face existing competitors or merely the threat of entry, and on whether firms compete primarily on the basis of price or long-run capacity decisions.

This literature is extensively reviewed in Verrecchia (2001) and Dye (2001). Unlike the previous five hypotheses on voluntary disclosure, the proprietary cost hypothesis assumes there are no conflicts of interest between managers and shareholders. As a result, this literature predicts that voluntary disclosure will always be credible. The focus of this literature, therefore, is on examining the economic forces that constrain full disclosure.

Hayes and Lundholm (1996) argue that proprietary costs induce firms to provide disaggregated data only when they have similarly-performing business segments. Firms with widely-varying performance across business segments have incentives to conceal these performance differences from competitors by only reporting aggregate performance.

There has been relatively little direct evidence on the proprietary cost hypothesis. Piotroski (1999a) examines firms' decisions to provide additional segment disclosures. He concludes that firms with declining profitability and with less variability in profitability across industry segments are more likely to increase segment disclosures, consistent with the proprietary cost hypothesis.

The proprietary cost hypothesis can be potentially extended to include other externalities from information disclosure. For example, Watts and Zimmerman (1986) argue that firms are concerned about potential political and contracting costs from financial disclosures, which may in turn affect their voluntary disclosure.

Credibility of Voluntary Dislcosure

The extent to which voluntary disclosure mitigates resource misallocation in the capital market depends on the degree of credibility of information on the firm's economics that is not available from other sources, including required disclosures. Because managers have incentives to make self-serving voluntary disclosures, it is unclear whether management disclosures are credible.

There are potentially two mechanisms for increasing the credibility of voluntary disclosures. First, third-party intermediaries can provide assurance about the quality of management's disclosures. Second, there can be validation of prior voluntary disclosures through required financial reporting itself.[11] For example, managers' forecasts of revenues

[11] Lundholm (1999) points out that this role of accounting can be exploited to increase disclosures on intangible assets. However, if the legal system cannot distinguish between random forecast errors from deliberate management bias, such disclosures can potentially impose significant litigation costs.

and earnings can be verified using actual realizations. This mechanism will be effective in making disclosures credible if there are adequate penalties for managers that knowingly make disclosures that are subsequently proven false. The legal system and board monitoring play an important role in imposing such penalties.

Much of the evidence on the credibility of voluntary disclosures focuses on the accuracy and stock price effects of management forecasts. Waymire (1984) and Ajinkya and Gift (1984) show that there are positive stock price reactions to management forecasts of earnings increases, and negative reactions to forecasts of earnings decreases.[12] Pownall and Waymire (1989) find that the market reaction to unexpected management earnings forecasts is similar in magnitude to the reaction to unexpected earnings announcements themselves. This suggests that management forecasts have comparable credibility to audited financial information.

There is also evidence that investors are justified in viewing management forecasts as providing credible new information. Tests of the accuracy of these forecasts indicate that they are more accurate than contemporaneous analysts' forecasts (see Hassell and Jennings (1986) and Waymire (1986)), and are unbiased (see McNichols (1989)). In addition, financial analysts appear to revise their forecasts in response to information reflected in management's forecasts (see Hassell, Jennings, and Lasser (1988)). Piotrowksi (1999a) provides evidence that voluntary disclosures other than management forecasts are also credible. He examines a sample of firms that increase segment reporting disclosures,

[12] Hutton, Miller and Skinner (2000) find that good news forecasts, however, are only informative if they are accompanied by verifiable forward-looking statements.

and finds that the expanded disclosure is associated with an increase in analysts' forecast accuracy and a decline in dispersion.

Limitations of studies on voluntary disclosure

One of the major limitations of the above studies is the difficulty in measuring the extent of voluntary disclosure. Researchers have used several proxies for this variable, including management forecasts [see Miller and Piotroski (2000)], and metrics based on the AIMR database [see Lang and Lundholm (1994 and 1997) and Healy, Hutton, and Palepu (1999)], and self-constructed measures (see Botosan (1997) and Miller (1999)]. However, each approach has its limitations.

There are some significant advantages to using management forecasts as a voluntary disclosure proxy. First, they can be precisely measured. Managers' estimates are typically either point or range estimates for earnings or revenues. Second, the timing of the disclosure is typically known. As a result it is possible to assess whether the forecast preceded or lagged particular changes in variables of interest using daily or weekly data. This enables researchers to conduct more powerful tests of motivations for and consequences of voluntary disclosure.

However, one limitation of management forecasts as a proxy is that their accuracy can be easily verified by outside investors through actual earnings realizations. In contrast, it is more difficult to ex post verify the accuracy of many other types of voluntary disclosures, such as customer satisfaction and human capital. As a result, research using management forecasts as the metric for voluntary disclosure is likely to

increase the power of the tests, but these findings may not generalize to other forms of voluntary disclosure.

The AIMR data provides a more general measure of voluntary disclosure than management forecasts. The annual survey produces firm rankings of aggregate voluntary disclosure for each industry covered in the survey, as well as disaggregate rankings for voluntary disclosure published in annual financial statements and 10-Ks, voluntary disclosures published in quarterly financial reports, and voluntary disclosure provided through firms' investor relations. The panels that provide the rankings comprise the leading analysts in each industry, and are therefore likely to be particularly well qualified to judge firms' disclosures. In addition, the metric covers all disclosure, including that through analyst meetings and conference calls. However, it is unclear whether the analysts on the AIMR panels take the ratings seriously, how they select firms be included in the ratings, and what biases they bring to the ratings.

Studies with self-constructed measures of disclosure face a different set of problems. Because the authors have developed their own metric of voluntary disclosure, there is increased confidence that the measure truly captures what is intended. However, to the extent that construction of the metrics involves judgment on the part of the researcher, the findings may be difficult to replicate. In addition, these metrics typically rely on disclosures provided in the annual report or other such public documents. As a result any disclosures that firms provide in analysts meetings, conference calls, and other such venues are omitted from the analysis.

Finally, endogeneity is a potentially serious problem for some of the above studies. For example, firms that have public capital market transactions are also likely to be facing changes in their investment opportunity sets. It is then difficult to assess whether the relation between high levels of disclosure and increases in disclosure for these firms is attributable to the public issue per se, or to other changes that the firm is experiencing.

In summary, the analyst ratings and self-constructed proxies are likely to be noisy measure of disclosure. This is likely to reduce the power of the tests used in examining the motives for voluntary disclosure.[13]

3.4 Capital Market Consequences of Reporting and Disclosure

Both the positive accounting theory and the voluntary disclosure literatures have examined the capital market consequences of changes in corporate reporting. Positive accounting theory research has focused on effects of changes in accounting methods and regulatory decisions to change standards. Voluntary disclosure research has examined the capital market effects of changes in corporate disclosure.

3.4.1 Positive Accounting Theory Literature

As noted above, the major focus of the positive accounting literature has been to document contracting and political factors that explain management financial reporting

[13] In contrast, the measurement problems that we discussed for the positive accounting literature are for the independent variables, leading to a concern about bias in the findings due to correlated omitted variables.

decisions. However, several studies have examined the economic consequences and shareholder wealth effects of changes in accounting choices. For example, studies of the effects of changes in oil and gas accounting standards find that firms required to change from the full cost method to successful efforts experienced a decline in stock prices (see Dyckman and Smith (1979) and Collins, Rozeff, and Dhaliwal (1981). There is some evidence that the decline is correlated with contracting variables (see Lys (1984)). However, studies of the stock price effects of other accounting standards report largely insignificant incremental contracting or political cost effects (Leftwich 1981). Similarly, studies of firms' accounting method changes indicate that there is generally no significant relation between stock returns at announcement of the accounting change and contracting or political cost considerations (see Holthausen (1981)).

There are at least three potential explanations for this finding. First, accounting decisions have no significant shareholder wealth effects. However, there is some evidence that contradicts this explanation. The oil and gas studies themselves find that there is a significant stock price effect associated with the unanticipated change in accounting standard. Also, Foster (1979) documents significant stock price changes for critiques of firms' accounting decisions by analysts such as Abraham Briloff.[14]

A second explanation is that it is difficult to measure stock price effects for many of the events studied (see Holthusen and Leftwich (1983) and Watts and Zimmerman

[14] One potential explanation for the market reaction is that Briloff's analysis provides new information about the validity of managers' forecasts underlying their accounting judgments. Some researchers argue that the "Briloff effect" reflects changes in the firms' economics subsequent to the publication of his articles, such as tax, litigation and regulatory effects. Foster examines this explanation, and concludes that the non-information related economic factors cannot fully explain the observed market reaction to the accounting critiques.

(1993)). For example, accounting standards are set through a lengthy process, making it difficult to capture shareholder wealth effects of the announcement of the standard itself. Similarly, it is often difficult to identify the date when outside investors first learned of an accounting change. Finally, since accounting changes are often accompanied by other economic changes, it is difficult to isolate the stock price effect of the accounting change itself.

A third explanation for the findings is that contracting and political cost considerations are economically unimportant in explaining wealth effects of reporting changes. Consistent with this explanation, Healy and Palepu (1991) find that the average annual CEO compensation effects from a change in depreciation accounting method amount to 1.5 percent of their base salary. Since CEO base salary is usually only a small fraction of a firm's market value, this evidence suggests that the CEO compensation effect of depreciation changes on shareholder wealth is likely to be insignificant.[15]

3.4.2 Voluntary Disclosure Literature

A number of studies examine the economic consequences of voluntary disclosure. These studies argue that there are potentially three types of capital market affects for firms that make extensive voluntary disclosures: improved liquidity for their stock in the capital market, reductions in their cost of capital, and increased following by financial analysts. Each of these effects and the relevant empirical evidence is discussed below. Since many of the studies discussed have common limitations, we discussed these

[15] One explanation for the small economic magnitude of compensation and other contracting effects is that investors anticipate such potential costs in writing the contracts. Surviving contracts, therefore, are likely to be efficient, making the observed contracting costs small. However, this does not suggest that the concept of contracting costs is unimportant for managers' accounting decisions.

limitations together, subsequent to the presentation of their main findings; the findings summarized below, therefore, should be interpreted with these limitations in mind.

1. Improved Stock Liquidity

Diamond and Verrecchia (1991), and Kim and Verrecchia (1994) argue that voluntary disclosure reduces information asymmetries among informed and uninformed investors. As a result, for firms with high levels of disclosure, investors can be relatively confident that any stock transactions occur at a "fair price," increasing liquidity in the firm's stock. In addition, these studies argue that expanded disclosure and stock liquidity will be associated with increased institutional ownership.

Several papers provide evidence that is consistent with this hypothesis. Healy, Hutton, and Palepu (1999) find that firms that expand disclosure experience significant contemporaneous increases in stock that is unrelated to current earnings performance. Gelb and Zarowin (2000) find that firms with high disclosure ratings have high stock price associations with contemporaneous and future earnings relative to firms with low disclosure ratings. These findings suggest that firms' disclosure strategies affect the speed with which information gets into prices.

In addition, several studies attempt to measure stock liquidity and to examine its relation to firm disclosure proxies. Welker (1995) documents a significant negative relation between analysts' ratings of firms' disclosures and bid-ask spreads. Healy, Hutton, and Palepu (1999) find firms with increased analyst ratings of disclosure had significantly higher bid-ask spreads than their industries prior to the disclosure change. After the disclosure increase, bid-ask spreads for the sample firms reverted to the same levels as their industry peers. Finally, Leuz and Verrecchia (2000) examine bid-ask spreads for firms listed on the Neuer Market, which has higher disclosure requirements. They find that these firms have lower bid-ask spreads than firms listed on the Frankfurt Exchange.

2. Reduced Cost of Capital

As discussed in Section 2, the lemons problem in capital markets creates an incentive for managers to provide voluntary disclosure to reduce the cost of capital. A similar argument is made by Barry and Brown (1984, 1985, 1986), who note that when disclosure is imperfect, investors bear risks in forecasting the future payoffs from their investment. If this risk is non-diversifiable, investors will demand an incremental return for bearing the information risk. As a result, firms with high levels of disclosure, and hence low information risk, are likely to have a lower cost of capital than firms with low disclosure levels and high information risk.

Botosan (1997) provides some evidence consistent with the cost of capital hypothesis. She finds that for firms with low analyst following, there is a negative relation between cost of equity capital and the extent of their voluntary disclosures. Piotroski (1999b) finds that firms providing additional segment disclosures have a contemporaneous increase in the market's capitalization of their earnings, consistent with the firm having a lower cost of capital.[16] Finally, Botosan and Plumlee (2000) find a negative cross-sectional relation between cost of capital and analyst rankings of annual report disclosures. However, they also find that firms' cost of capital is positively related to rankings of quarterly disclosures, and unassociated with investor relations activities.

3. Increased Information Intermediation

Bhushan (1989a, 1989b) and Lang and Lundholm (1999) argue that if management's private information is not fully revealed through required disclosures, voluntary disclosure lowers the cost of information acquisition for analysts and hence increases their supply. However, the effect of voluntary disclosure on the demand for analysts' services is ambiguous. Expanded disclosure enables financial analysts to create

[16] The increased segment disclosure is potentially endogenous to the changes in the economics of the sample firms, potentially confounding this conclusion. The endogeneity issue is a common concern for many other studies discussed in this section. We discuss this issue further later.

valuable new information, such as superior forecasts and buy/sell recommendations, thereby increasing demand for their services. However, public voluntary disclosure also pre-empts analysts' ability to distribute managers' private information to investors, leading to a decline in demand for their services.

Lang and Lundholm (1993) find that firms with more informative disclosures have larger analyst following, less dispersion in analyst forecasts, and less volatility in forecast revisions. Healy, Palepu and Hutton (1999) show that firms with increased analyst ratings of disclosure have lower analyst coverage than their industry peers in the pre-event period. After the increase in disclosure, however, analyst coverage for the sample firms reverts to the same level as other firms in the industry. Finally, Francis, Hanna and Philbrick (1998) find that there is an increase in analyst coverage for firms making conference calls.

Limitations of Studies of Voluntary Disclosure Capital Market Consequences

Potential endogeneity is the most important limitation of the above findings. For example, firms with the highest disclosure ratings tend to also show the highest contemporaneous earnings performance [see Lang and Lundholm (1993)]. This may be caused by a self-selection bias – firms may increase disclosure when they are performing well. As a result the association between capital market variables and disclosure may be driven by firm performance rather than disclosure per se. More generally, disclosure changes are unlikely to be random events: they are likely to coincide with changes in firm economics and governance.

Several studies attempt to control for performance changes to isolate the impact of disclosure. For example, Healy, Hutton, and Palepu (1999) control for contemporaneous

earnings levels and earnings changes in examining the cross-sectional relation between disclosure increases and variables such as stock performance, analyst following, institutional ownership, and analyst forecast dispersion. However, such controls are likely to be imperfect in the absence of a reliable model of the relation between performance and disclosure. In addition, as noted above, there are likely to be other correlated omitted variables in these analyses.

A related problem with both AIMR and self-constructed measures of voluntary disclosure is that it is difficult to precisely define the timing of any change in disclosure. Typically disclosure is measured for a given year, making it difficult to infer whether disclosure changes followed or preceded changes in variables of interest. Consequently, it is difficult to draw strong conclusions about the direction of causality underlying the documented associations.

4. Where do we go from here?

In this section we summarize what we have learnt from the disclosure literature and what questions remain unresolved. We then examine current trends in the economy and discuss the challenges and opportunities that they provide for future research.

The following broad conclusions are supported by empirical research discussed in this paper:

(i) Regulated financial reports are informative to investors, and the degree of informativeness varies systematically with firm and economy characteristics.

(ii) Financial analysts add value in the capital market through their analysis of firms' financial reporting decisions, forecasts of future earnings, and buy/sell recommendations.

(iii) There is a market-driven demand for auditing services.

(iv) Both financial analysts and auditors are imperfect intermediaries, in part because of incentive conflicts.

(v) Managers' financial reporting and disclosure choices are influenced by contracting, political cost and capital market considerations.

(vi) There is mixed evidence on the economic consequences of managers' financial reporting and disclosure choices.

Despite the progress in the last thirty years, many of the questions identified in table 1 have yet to be fully addressed, or are not yet answered. We have already discussed several specific ideas for future research throughout Section 3. We believe that recent macro-economic forces create several additional opportunities future research. We discuss four forces: rapid technological innovations, the advent of network organizations, changes in business economics of audit firms and financial analysts, and globalization.

(i) Rapid Technological Innovation

There have been phenomenal technological innovations in the last twenty years in areas such as computers, communications, biotechnology, and the internet. The economic consequences of these innovations are typically not reflected in financial statements in a timely manner. As noted earlier, except for software R&D occurring after technological

feasibility, U.S. firms expense R&D outlays immediately, regardless of their economic values. As a result, investors that are interested in assessing the potential economic performance of innovative firms in the current period, as well as potential future benefits from innovations-in-progress are forced to look beyond the financial statements.

Chang (1998) and Lev and Zarowin (1999) find that the decline in value relevance of financial statement items is partially explained by an increase in innovation. Further, Amir and Lev (1996) show that for firms in the wireless communications industry nonfinancial indicators of performance, such as market population size (POPS) and market penetration, have a more significant relation to stock prices than financial statement information.

Technological innovation has also created new channels for investor communication. For example, conference calls and the internet make it easier for firms to communicate rapidly with key investors and financial intermediaries. Conference calls are large-scale telephone conversations between managers and key financial analysts, where managers provide voluntary disclosure by answering analysts' questions about the firm's current and future performance. Tasker (1998) documented that 35% of mid-sized firms hosted a conference call in the period 1995 to 1996 and that many firms used this channel to mitigate limitations in required financial reporting.

The internet provides management with the opportunity to access all investors and to provide daily updates of important information. Many corporate internet sites provide an overview of the company's performance, a review of performance, press releases, stock quotes, frequently asked investor relations questions, earnings forecasts

(by financial analysts), as well as annual reports, SEC reports. The increasing use of the internet by investors is likely to continue, reducing the costs of providing voluntary disclosures and presumably increasing their supply.

(ii) Network Organizations

Innovations in organizational forms, such as closely-coordinated supply chains and strategic alliances, also significantly affect the nature of financial reporting and disclosure. These organizational forms facilitate efficient risk sharing and market-based coordination of activities that were traditionally performed internally. However, by blurring the boundaries of firms, they pose difficult challenges to the entity measurement concept. For example, it is difficult for financial statements to fully reflect the complex relations and implicit commitments that underlie network relations between Coca Cola Company and its bottlers. As a result of its exclusive contracts with bottlers, Coke has been able to outsource the capital-intensive, low margin activities for its business. Consequently, bottlers have reported marginal financial performance while Coke has shown strong earnings performance. However, the reported financial performance of Coke does not fully reflect the complex relationship and implicit commitments between the companies. Reflecting these types of interdependencies in financial statements is challenging for standard setters. Current standards ignore them, potentially reducing the timeliness of accounting information.

(iii) Changes in Business Economics of Audit Firms and Financial Analysts

A third current phenomenon is changing business economics of audit firms and financial analysts. Audit firms are increasingly reliant on management consulting, rather than assurance services. Regulators and commentators have argued that there is a potential conflict between the two. For example, they question whether an audit firm will challenge management of corporate clients if the same management is responsible for hiring the audit firm for a consulting engagement. To our knowledge there is no large sample empirical evidence of a decline in the credibility of audit reports. However, the issue has prompted the SEC to propose that audit firms divest their consulting practices.

There have also been changes in the business model of financial analysts. The decline in trading costs has reduced soft dollars available for funding research. Therefore, these activities are increasingly linked to investment banking and underwriting activities. In addition, leading financial analysts are beginning to be viewed as strategy advisors to the companies that they are analyzing. For example, the financial press is replete with stories on the increasingly important roles of telecommunication analyst Jack Grubman and internet analyst Mary Meeker in the strategic decisions of their respective industries (see Business Week, 2000). These trends are likely to exacerbate potential conflicts of interest that analysts face, raising questions about the effectiveness as information intermediaries.

(iv) Globalization

Capital markets are becoming increasingly global as a result of a variety of developments. Institutional investors are looking to diversify by investing around the globe; corporations are seeking capital wherever the terms are most attractive; and internet-based trading is making it easier for individual investors to invest in international capital markets. Financial deregulation is encouraging these activities.

The globalization of capital markets has been accompanied by calls for globalization of financial reporting. This raises several interesting questions. First, is it optimal to have a global accounting standard setter given wide disparities in the development of financial reporting infrastructure across counties? Second, what economic forces will determine the speed with which convergence of financial reporting institutions will take place? Third, what are the political and economic consequences of such a convergence? Fourth, in the absence of convergence, will financial reporting informativeness be enhanced by global accounting standards?

In summary, the increased pace of entrepreneurship and economic change has probably increased the value of reliable information in capital markets. However, the traditional financial reporting model appears to do a poor job of capturing the economic implications of many of these changes in a timely way. There is, therefore, an opportunity for future disclosure research to examine how financial reporting and disclosures adapt to changes in business and capital market environments. In addition, as we note earlier, there are many areas where our understanding of existing disclosure

institutions and phenomena are limited. We believe that both opportunities make the disclosure area an exciting area of study for accounting scholars.

Figure 1

Financial and Information Flows in a Capital Market Economy

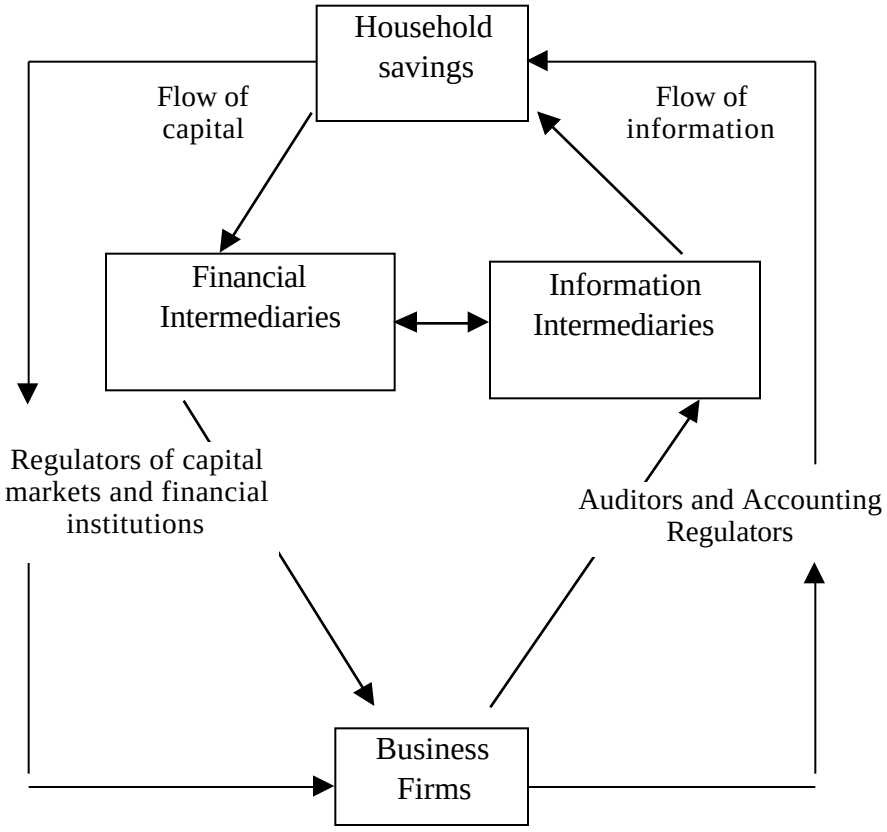


Table 1

Research questions implied by disclosure framework

Topic	Questions
Regulation of disclosure.	· Why is there a need for regulation of disclosure in capital markets? What types of disclosures should be regulated and which should not? · How effective are accounting standards in facilitating credible communication between managers and outside investors? What factors determine their effectiveness? · Which mandated disclosures should be recognized directly in the financial statements and which should be included as supplemental disclosures?
Auditing/intermediaries and disclosure	· How effective are auditors in enhancing the credibility of financial statements? What factors influence auditors' effectiveness? · How effective are financial analysts as information intermediaries? What factors influence their effectiveness? · How does corporate disclosure affect analyst coverage of firms?
Managers disclosure decisions	· What factors affect management's disclosure choices? · What is the relation between disclosure, corporate governance, and management incentives? What role do boards and audit committees play in the disclosure process?

Capital market consequences of disclosure	· How do investors respond to corporate disclosures? Are firm disclosures made outside the financial statements credible? · Do investors evaluate disclosures that are included directly in the financial statements differently from those that are included as supplemental disclosures? · What factors influence investors' perception of the quality of capital market disclosures across economies? · How does disclosure affect resource allocation in the economy?

REFERENCES

Abrahamson, E. and E. Amir, 1996, The information content of the president's letter to shareholders, Journal of Business Finance & Accounting 23, 1157-1183.

Ajinkya, B. and M. Gift, 1984, Corporate managers' earnings forecasts and symmetrical adjustments of market expectations, Journal of Accounting Research 22, 425-44.

Akerlof, G., 1970, The market for 'lemons': Quality uncertainty and the market mechanism, Quarterly Journal of Economics 90, 629-650.

Alford, A., J. Jones, R. Leftwich, and M. Zmijewski, 1993, The relative informativeness of accounting disclosures in different countries, Journal of Accounting Research 31, 183-224.

Amir, E., 1993, The market valuation of accounting information: The case of postretirement benefits other than pensions, The Accounting Review 68, 703-725.

Amir, E. and B. Lev, 1996, Value-relevance of nonfinancial information: The wireless communications industry, Journal of Accounting & Economics 22, 3-30.

Ball, R., S.P. Kothari, and A. Robin, 2000, The effect of international institutional factors on properties of accounting earnings, Journal of Accounting & Economics, forthcoming.

Ball, R., A. Robin and J. Wu, 2000, Incentives versus standards: Properties of accounting income in four East Asian countries, and implications for acceptance of IAS, Working Paper, University of Rochester (Rochester, NY).

Barry, C.B. and S.J. Brown, 1984, Differential information and the small firm effect**,** Journal of Financial Economics 13: 2, 283-295.

Barry, C.B. and S.J. Brown, 1985, Differential information and security market equilibrium, Journal of Financial and Quantitative Analysis 20, 407-422.

Barry, C.B. and S. J. Brown, 1986, Limited information as a source of risk, The Journal of Portfolio Management, 66-72.

Barth, M.E., 1991, Relative measurement errors among alternative pension asset and liability measures, The Accounting Review 66: 3, 433-464.

Barth, M.E. and A.P. Hutton, 2000, Information intermediaries and the pricing of accruals, Working Paper, Stanford University (Stanford, CA).

Barth, M., W. Beaver and W. Landsman, 2001, Comments on the relevance of the value-relevance literature for financial accounting standard setting, Journal of Accounting and

Economics, Conference Volume, forthcoming.

Beaver, W., A. Christie and P. Griffin, 1980, The information content of SEC accounting series release No. 190, Journal of Accounting and Economics 2, 127-157.

Beaver, W. and M. McNichols, 1998, The characteristics and valuation of loss reserves of property casualty insurers, Review of Accounting Studies 3: 1-2.

Beaver, W., 1998, Financial reporting: An accounting revolution, Third Edition, Prentice-Hall (Englewood Cliffs, NJ).

Beneish, M.D., 1999, Incentives and penalties related to earnings overstatements that violate GAAP, The Accounting Review 74, 425-457.

Bhushan, R., 1989a, Collection of information about publicly traded firms: Theory and evidence, Journal of Accounting & Economics 11, 183-207.

Bhushan, R., 1989b, Firm characteristics and analyst following, Journal of Accounting & Economics 11: 2,3, 255-275.

Botosan, C.A., 1997, Disclosure level and the cost of equity capital, The Accounting Review 72: 3, 323-350.

Botosan, C. A. and M. A. Plumlee, 2000, A re-examination of disclosure level and expected cost of capital, Unpublished working paper, University of Utah.

Brennan, N., 1999, Voluntary disclosure of profit forecasts by target companies in takeover bids, Journal of Business Finance & Accounting 26, 883-918.

Brown, L., G. Foster and E. Noreen, 1985, Security analyst multi-year earnings forecasts and the capital market. Studies in Accounting Research, No. 23, American Accounting Association (Sarasota, FL).

Brown, L. and M. Rozeff, 1978, The superiority of analyst forecasts as measures of expectations: Evidence from earnings, Journal of Finance 33, 1-16.

Brown, L., 1997, Analyst forecasting errors: Additional evidence, Financial Analysts' Journal, Nov/Dec, 81-88.

Brown, L., P. Griffin, R. Hagerman, and M. Zmijewski, 1987, Security analyst superiority relative to univariate time-series models in forecasting quarterly earnings, Journal of Accounting and Economics 9, 61-87.

Brown, S., K. Lo and T. Lys, 1998, Use of R^2 in accounting research: Measuring changes in value-relevance over the last four decades, Working paper, Northwestern University.

Chang, J., 1998, The decline in value relevance of earnings and book values, Working paper, University of Pennsylvania (Philadelphia, PA).

Choi, S.K. and D.C. Jeter, 1992, The effects of qualified audit opinions on earnings response coefficients, Journal of Accounting & Economics 15, 229-248.

Christie, A. and J. Zimmerman, 1994, Efficient and opportunistic choices of accounting procedures, The Accounting Review 69, 539-67.

Collins, D., E. Maydew and I. Weiss, 1997, Changes in the value-relevance of earnings and book values over the past forty years, Journal of Accounting & Economics 24: 1, 39-68.

Collins, D. and S. Kothari, 1989, An analysis of interpemproal and cross-sectional determinants of earnings response coefficients, Journal of Accounting & Economics 11: 2,3, 143-82.

Collins, D., M. Rozeff and D. Dhaliwal, 1981, The economic determinants of the market reaction to proposed mandatory accounting changes in the oil and gas industry, Journal of Accounting & Economics 3: 1, 37-72.

Darrough, M. and N. Stoughton, 1990, Financial disclosure policy in an entry game, Journal of Accounting & Economics 12, 219-244.

Darrough, M., 1993, Disclosure policy and competition: Cournot vs. Bertrand, The Accounting Review 68, 534-562.

DeAngelo, L., 1988, Managerial competition, information costs, and corporate governance: The use of accounting performance measures in proxy contests, Journal of Accounting & Economics 10, 3-37.

DeAngelo, H., L. DeAngelo, and D.J. Skinner, 1996, Reversal of fortune: Dividend signaling and the disappearance of sustained earnings growth, Journal of Financial Economics 40: 3, 341-372.

Dechow, P., A. Hutton, and R. Sloan, 1996, Causes and consequences of earnings manipulation: An analysis of firms subject to enforcement actions by the SEC, Contemporary Accounting Research 13, 1-36.

Dechow, P., A. Hutton and R. Sloan, 2000, The relation between analysts' forecasts of long-term earnings growth and stock price performance following equity offerings, Contemporary Accounting Research 17:1, 1-32.

Dechow, P., 1994, Accounting earnings and cash flows as measures of firm performance: The role of accounting accruals, Journal of Accounting & Economics 18, 3-43.

Dechow, P., S.P. Kothari, and R. Watts, 1998, The relation between earnings and cash flows, Journal of Accounting & Economics 25: 2, 133-168.

DeFond, M., T.J. Wong and S. Li, 1999, The impact of improved auditor independence on audit market concentration in China, Journal of Accounting & Economics 28: 3, 269-305.

Diamond, D. and R. Verrecchia, 1991, Disclosure, liquidity, and the cost of capital, The Journal of Finance 66, 1325-1355.

Dodd, P., N. Dopuch, R. Holthausen, and R. Leftwich, 1984, Qualified audit opinions and stock prices: Information content, announcement dates, and concurrent disclosures, Journal of Accounting & Economics 6: 1, 3-39.

Dodd, P., R. Holthausen and R. Leftwich, 1986, Abnormal stock returns associated with media disclosures of ‘subject to’ qualified audit opinions, Journal of Accounting & Economics 8: 2, 93-118.

Dopuch, N., R. Holthausen, and R. Leftwich, 1986, Abnormal stock returns associated with media disclosures of 'subject to' qualified audit opinions, Journal of Accounting & Economics 8, 93-118.

Dopuch, N., R. Holthausen, and R. Leftwich, 1987, Predicting audit qualifications with financial and market variables, The Accounting Review 62, 431.

Dyckman, T. and A. Smith, 1979, Financial accounting and reporting by oil and gas producing companies: A study of information effects, Journal of Accounting & Economics 1, 45-75.

Dye, R., 1983, Communication and post-decision information, Journal of Accounting Research 21, 514-534.

Dye, R., 2001, Commentary on essays on disclosure, Journal of Accounting and Economics, Conference Volume, forthcoming.

Easton, P. and M. Zmijewski, 1989, Cross-sectional variation in the stock market response to accounting earnings announcements, Journal of Accounting & Economics 11, 117-42.

Easton, P., T. Harris, and J. Ohlson, 1992, Aggregate accounting earnings can explain most of security returns: The case of long return intervals, Journal of Accounting & Economics 15: 2,3, 119-143.

Eccher, E. and P. Healy, 2000, The role of international accounting standards in transitional economies: A study of the People's Republic of China, Unpublished working paper, Harvard Business School.

Ely, K. and G. Waymire, 1999, Accounting standard-setting organizations and earnings relevance: Longitudinal evidence from NYSE common stocks, 1927-93, Journal of Accounting Research 37, 293-318.

Feltham, G. and J. Xie, 1992, Voluntary financial disclosure in an entry game with continua of type, Contemporary Accounting Research 9, 46-80.

Fields, T., T. Lys and L. Vincent, 2001, Empirical research on accounting choice, Journal of Accounting and Economics, Conference Volume, forthcoming.

Foster, G., 1979, Briloff and the capital market, Journal of Accounting Research 17: 1, 262-274.

Foster, G., 1987, Rambo IX: Briloff and the capital market, Journal of Accounting, Auditing and Finance 2, 409-431.

Francis, J., J. Hanna and D. Philbrick, 1998, Management communications with securities analysts, Journal of Accounting & Economics 24, 363-394

Francis, J., D. Philbrick and K. Schipper, 1994, Shareholder litigation and corporate disclosures, Journal of Accounting Research 32, 137-165.

Francis, J. and K. Schipper, 1998, Earnings surprises and litigation risk, Journal of Financial Statement Analysis 3, 15-28.

Francis, J. and K. Schipper, 1999, Have financial statements lost their relevance? Journal of Accounting Research 37, 319-53.

Francis, J. and L. Soffer, 1997, The relative informativeness of analysts' stock

recommendations and earnings forecast revisions**,** Journal of Accounting Research 35: 2, 193-212.

Frankel, R., M. McNichols, and P. Wilson, 1995, Discretionary disclosure and external financing, Accounting Review 70, 135-150.

Frankel, R., 1999, An empirical examination of conference calls as a voluntary disclosure medium, Journal of Accounting Research 37, 133-151.

Frost, C.A., 1997, Disclosure policy choices of UK firms receiving modified audit reports, Journal of Accounting & Economics 23, 163-188.

Gelb, D. and P. Zarowin, 2000, Corporate disclosure policy and the informativeness of stock prices, Working Paper, New York University (New York, NY).

Gheyara, K. and J. Boatsman, 1980, Market reaction to the 1976 replacement cost disclosures, Journal of Accounting & Economics 2, 107-125.

Gilson, S., P. Healy, C. Noe and K. Palepu, 2000, Conglomerate stock breakups and analyst specialization, Working Paper, Harvard Business School (Boston, MA).

Gigler, F., 1994, Self-enforcing voluntary disclosures, Journal of Accounting Research 32, 224-241.

Givoly, D., 1982, Financial analysts' forecasts of earnings: A better surrogate for market expectations, Journal of Accounting & Economics 4: 2, 85-108.

Givoly, D. and J. Lakonishok, 1979, The information content of financial analysts' forecasts of earnings: Some evidence on semi-strong efficiency, Journal of Accounting & Economics 2, 165-186.

Hand, J., R. Holthausen, and R. Leftwich, 1992, The effect of bond rating agency announcements on bond and stock prices, The Journal of Finance 47, 733-753.

Hassell, J., R. Jennings, and D. Lasser, 1988, Management earnings forecasts: Their usefulness as a source of firm-specific information to security analysts, The Journal of Financial Research 11, 303-320.

Hassell, J. and R. Jennings, 1986, Relative forecast accuracy and the timing of earnings forecast announcements, The Accounting Review 61, 58-76.

Hayes R. and R. Lundholm, 1996, Segment reporting to the capital market in the presence of a competitor, Journal of Accounting Research 34, 261-280.

Healy, P., A. Hutton and K. Palepu, 1999, Stock performance and intermediation changes surrounding sustained increases in disclosure, Contemporary Accounting Research 16, 485-520.

Healy, P., S. Kang and K. Palepu, 1987, The effect of accounting procedure changes on CEO's cash salary and bonus compensation, Journal of Accounting and Economics 9, 7-34.

Healy, P. and K. Palepu, 1990, Effectiveness of accounting-based dividend covenants, Journal of Accounting and Economics 12: 1-3, 97-124.

Healy, P. and K. Palepu, 1993, The effect of firms' financial disclosure strategies on stock prices, Accounting Horizons 7, 1-11.

Healy, P. and K. Palepu, 1995, The challenges of investor communications: The case of CUC International, Inc., Journal of Financial Economics 38, 111-141.

Healy, P., S. Myers and C. Howe, 1999, R&D accounting and the tradeoff between relevance and objectivity, Unpublished working paper, Harvard Business School.

Hirshleifer, J., 1971, The private and social-value of information and the reward to inventive activity**,** The American Economic Review 61: 4, 561

Holthausen, R., 1981, Evidence on the effect of bond covenants and management compensation contracts on the choice of accounting techniques: The case of the depreciation switch-back, Journal of Accounting and Economics 3: 1, 73-109.

Holthausen, R. and R. Leftwich, 1983, The economic consequences of accounting choice: Implications of costly contracting and monitoring, Journal of Accounting and Economics 5, 77-117.

Holthausen, R. and R. Leftwich, 1986, The effect of bond rating changes on common stock prices, Journal of Financial Economics 17, 57-90.

Holthausen, R. and R. Watts, 2001, The relevance of the value-relevance literature for financial accounting standard setting, Journal of Accounting and Economics, Conference Volume, forthcoming.

Hutton, A., G. Miller, and D. Skinner, 2000, Effective voluntary disclosure, Unpublished working paper, Harvard Business School.

Imhoff, E., 1978, The representativeness of management earnings forecasts, The Accounting Review 53, 836.

Jacob, J., T. Lys, M. Neale, 1999, Expertise in forecasting performance of security analysts, Journal of Accounting and Economics 28, 51-82.

Jennings, R., 1987, Unsystematic security price movements, management earnings forecasts, and revisions in consensus analyst earnings forecasts, Journal of Accounting Research 24, 90-110.

Kasznik, R., 1999, On the association between voluntary disclosure and earnings management, Journal of Accounting Research 37, 57-83.

Kim, O. and R. Verrecchia, 1994, Market liquidity and volume around earnings announcements, Journal of Accounting & Economics 17, 41-68.

Kothari, S.P., 2001, Capital markets research in accounting, Journal of Accounting and Economics, Conference Volume, forthcoming.

Kreps, D. 1990, A course in microeconomic theory, (Princeton University Press, Princeton, NJ).

Lambert, R., 2001, Contracting theory and accounting, Journal of Accounting and Economics, Conference Volume, forthcoming.

Lang, M. and R. Lundholm, 1993, Cross-sectional determinants of analysts ratings of corporate disclosures, Journal of Accounting Research 31, 246-271.

Lang, M. and R. Lundholm, 1997, Voluntary disclosure during equity offerings: Reducing information asymmetry or hyping the stock? Working Paper, University of Michigan (Ann Arbor, MI).

Lang, M. and R. Lundholm, 1999, Corporate disclosure policy and analyst behavior, The Accounting Review 71, 467-493.

Leftwich, R., 1980, Market failure fallacies and accounting information, Journal of Accounting & Economics 2, 193-211.

Leftwich, R., 1981, Evidence of the impact of mandatory changes in accounting principles on corporate loan agreements, Journal of Accounting & Economics 3: 1, 3-37.

Leftwich, R., 1983, Accounting Information in Private Markets: Evidence from Private Lending Agreements, The Accounting Review 58: 1, 23-43.

Leuz, C. and R. Verrecchia, 2000, The economic consequences of increased disclosure, Journal of Accounting Research, forthcoming.

Lev, B. and P. Zarowin, 1999, The boundaries of financial reporting and how to extend them, Journal of Accounting Research, 37, 353-386.

Lin, H. and M. McNichols, 1998, Underwriting relationships, analysts' earnings forecasts

and investment recommendations, Journal of Accounting & Economics 25: 1, 101-128.

Lundholm, R., 1999, Reporting on the past: A new approach to improving accounting today, Accounting Horizons 13, 315-323.

Lys, T., 1984, Mandated accounting changes and debt covenants: The case of oil and gas accounting, Journal of Accounting & Economics 6: 1, 39-66.

Lys, T and S. Sohn, 1990, The association between revisions of financial analysts' earnings forecasts and security price changes, Journal of Accounting & Economics 13, 341-364.

Matsumoto, D., 2000, Management's incentives to avoid negative earnings surprises, Working Paper, Harvard Business School (Boston, MA).

McNichols, M., 1989, Evidence of informational asymmetries from management earnings forecasts and stock returns, The Accounting Review 64, 1-27.

Merton, R.C., 1987, A simple model of capital market equilibrium with incomplete information, The Journal of Finance 42, 483-510.

Miller, G., 1999, Earnings performance and discretionary disclosure, Unpublished working paper, Harvard University.

Miller, G. and J. Piotroski, 2000, The role of disclosure for high book-to-market firms, Unpublished working paper, Harvard University.

Morck, R., A. Shleifer, and R. Vishny, 1990, Do managerial objectives drive bad acquisitions? Journal of Finance 45, 31-50.

Myers, S. and N. Majluf, 1984, Corporate financing and investment decisions when firms have information that investors do not have, Journal of Financial Economics 13, 187-222.

Newman, P. and R. Sansing, 1993, Disclosure policies with multiple users, Journal of Accounting Research 31, 92-113.

Noe, C., 1999, Voluntary disclosures and insider transactions, Journal of Accounting & Economics 27, 305-27.

Palepu, K., 1986, Predicting takeover targets: A methodological and empirical analysis, Journal of Accounting & Economics 8, 3-36.

Palepu, K., 1987, An anatomy of an accounting change, in Accounting and management: Field study perspectives, edited by W. Bruns Jr. and R. Kaplan, HBS Press (Boston MA).

Piotroski, J., 1999a, Discretionary segment reporting decisions and the precision of investor beliefs, Working paper, University of Chicago.

Piotroski, J., 1999b, The impact of reported segment information on market expectations and stock prices, Working paper, University of Chicago.

Posner, R., 1974, Theories of economic regulation, Bell Journal of Economics and Management Science 5, 335-358.

Pownall, G., C. Wasley, and G. Waymire, 1993, The stock price effects of alternative types of management earnings forecasts, The Accounting Review 68, 896-912.

Pownall, G. and G. Waymire, 1989, Voluntary disclosure credibility and securities prices: Evidence from management earnings forecasts, Journal of Accounting Research 27, 227- 246.

Ro, B., 1980, The adjustment of security returns to the disclosure of replacement cost accounting information, Journal of Accounting & Economics 2: 159-89.

Sansing, R., 1992, Accounting and the credibility of management forecasts Contemporary Accounting Research 9, 33-46.

Skinner, D., 1993, The investment opportunity set and accounting procedure choice: Preliminary evidence, Journal of Accounting & Economics 17, 407-446.

Skinner, D., 1994, Why firms voluntarily disclose bad news, Journal of Accounting Research 32, 38-61.

Skinner, D., 1997, Earnings disclosures and stockholder lawsuits, Journal of Accounting & Economics, 23, 249-283.

Sloan, R., 1990, Accounting earnings and top executive compensation, Journal of Accounting & Economics 16: 1-3, 55-101.

Smith, C., and R. Watts, 1992, The investment opportunity set and corporate financing, dividend, and compensation policies, Journal of Financial Economics 32, 263-292.

Tasker, S., 1998, Bridging the information gap: Quarterly conference calls as a medium for voluntary disclosure, Review of Accounting Studies 3.

Trueman, B., 1986, Why do managers voluntarily release earnings forecasts? Journal of Accounting & Economics 8, 53-72.

Verrecchia, R., 1983, Discretionary disclosure, Journal of Accounting & Economics 5, 179-194.

Verrecchia, R., 1990, Information quality and discretionary disclosure. Journal of Accounting & Economics 12, 365-380.

Verrecchia, R, 2001, Essays on disclosure, Journal of Accounting and Economics, Conference Volume, forthcoming.

Wagenhofer, A. 1990 Voluntary disclosure with a strategic opponent. Journal of Accounting and Economics 12, 341-364.

Warner, J., R. Watts, and K. Wruck, 1988, Stock prices and top management changes, Journal of Financial Economics 20, 461-493.

Watts, R. and J. Zimmerman, 1978, Towards a positive theory of the determination of accounting standards, The Accounting Review 53: 1, 112-134.

Watts, R. and J. Zimmerman, 1981a, The markets for independence and independence auditors, Working Paper, University of Rochester (Rochester, NY).

Watts, R. and J. Zimmerman, 1981b, Auditors and the determination of accounting standards, Working Paper, University of Rochester (Rochester, NY).

Watts, R. and J. Zimmerman, 1986, Positive accounting theory, Prentice-Hall (Englewood Cliffs, NJ).

Watts, R. and J. Zimmerman, 1990, Positive accounting theory: A ten year perspective, The Accounting Review 65, 131-157.

Watts, R. and J. Zimmerman, 1983, Agency problems, auditing and the theory of the firm: Some evidence, Journal of Law & Economics 26, 613-34.

Waymire, G., 1984, Additional evidence on the information content of management earnings forecasts, Journal of Accounting Research 22, 703-719.

Waymire, G., 1986, Additional evidence on the accuracy of analyst forecasts before and after voluntary management earnings forecasts, The Accounting Review 61, 129-143.

Waymire, G., 1985, Earnings volatility and voluntary management forecast disclosure, Journal of Accounting Research 23, 268.

Welker, M., 1995, Disclosure policy, information asymmetry and liquidity in equity markets, Contemporary Accounting Research 11, 801-828.

Weisbach, M., 1988, Outside directors and CEO turnover, Journal of Financial Economics 20, 431-461.

Zimmer, I., 1986, Accounting for interest by real estate developers, Journal of Accounting & Economics 8, 37-52.